ALPAS: A SIMULATION-BASED DRRM EVACUATION PLANNING SYSTEM WITH OPTIMIZED ROUTING AND QUEUE MANAGEMENT

A Project Proposal Submitted to the Polytechnic University of the Philippines – Parañaque City Campus
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COMP 014 Quantitative Methods with Modeling and Simulation

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I. PROJECT OVERVIEW AND OBJECTIVES

1.1 Overview

Disasters such as earthquakes, fires, and other emergencies remain unpredictable yet inevitable, particularly in densely populated institutions like schools and universities. In the Philippines, disaster preparedness is emphasized by the National Disaster Risk Reduction and Management Council (NDRRMC), highlighting the need for efficient evacuation planning and response systems. Traditional evacuation plans, often limited to static maps and periodic drills, fail to account for dynamic variables such as crowd density, panic behavior, and route congestion. This study introduces **ALPAS** (Advanced Learning Platform for Assessing Safety), is a professional-grade fire and evacuation simulation platform developed as the school-fire-sim web application. The system combines agent-based evacuation modeling, fire dynamics principles (heat release rate growth, visibility, temperature, and carbon monoxide), and interactive 2D/3D visualization to help students, faculty, researchers, engineers, and safety professionals evaluate school building scenarios before real emergencies occur.

1.2 Objectives

General Objective:

To develop and evaluate a simulation-based DRRM evacuation planning system that improves evacuation efficiency through queueing and path optimization techniques.

Specific Objectives:

- To design a simulation model that represents crowd movement and congestion using queueing theory.

- To implement a path optimization algorithm that determines efficient evacuation routes.
- To assess the effectiveness of the system in reducing evacuation time and congestion.
- To integrate essential emergency information, including hotlines and nearby evacuation areas, into the system.
- Provide clarity: replace panic with understanding through rigorous simulation of fire dynamics and human behavior.
- Build courage: enable preparation through repeatable what-if scenarios and mitigation comparisons.
- Restore control: support better design, training, and mitigation strategies via measurable ASET/RSET margins.
- Deliver dual fidelity: Quick client-side analysis for teaching and demos; FDS+JuPedSim runs for research-grade artifacts.
- Integrate AI-assisted interpretation of run logs and metrics through the AI Analyst module.

1.3 Scope and Limitations

The scope of this study is limited to academic institutions, specifically at Polytechnic University of the Philippines — Parañaque City Campus. The system focuses on simulating evacuation scenarios during disasters such as earthquakes and fire disasters, applying queueing theory and path optimization to improve evacuation efficiency. The study does not cover large-scale community evacuations or external disaster response operations. Limitations include reliance on simulated data rather than live disaster conditions, potential constraints in integrating the system with existing school infrastructure, and the need for continuous updates to reflect evolving disaster management protocols.

1.4 Target Audience and Stakeholders

The primary target audience includes students and school administrators who will directly benefit from improved evacuation guidance and preparedness. Stakeholders also include disaster risk reduction practitioners, local government units (LGUs), and future researchers.

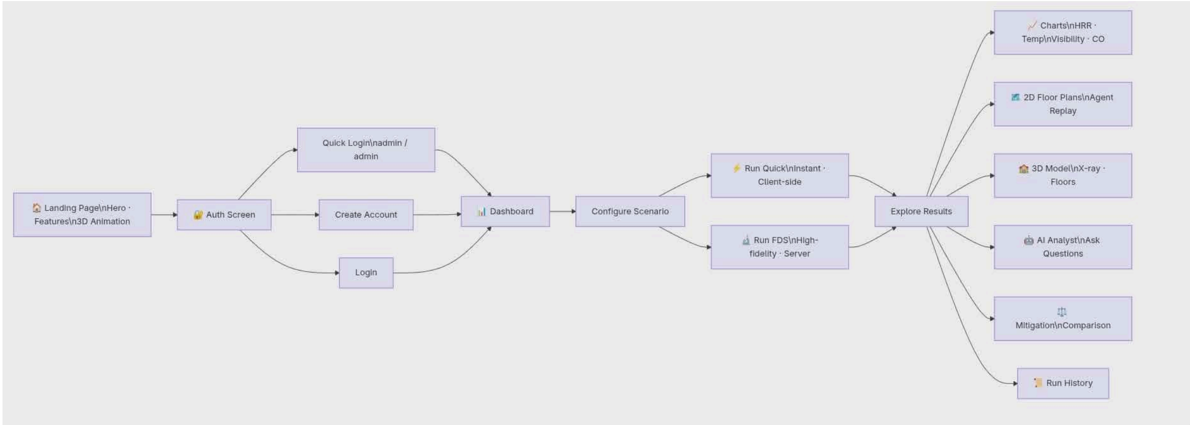
- **Students:** Gain awareness and preparedness through interactive evacuation guidance.
- **School Administrators:** Obtain data-driven insights for improving evacuation plans.
- **Disaster Risk Reduction Practitioners:** Align evacuation planning with national frameworks.
- **Local Government Units (LGUs):** Enhance community-level disaster preparedness strategies.
- **Future Researchers:** Build upon the foundation for further studies in simulation systems and DRRM innovations.

II. MODEL DESIGN

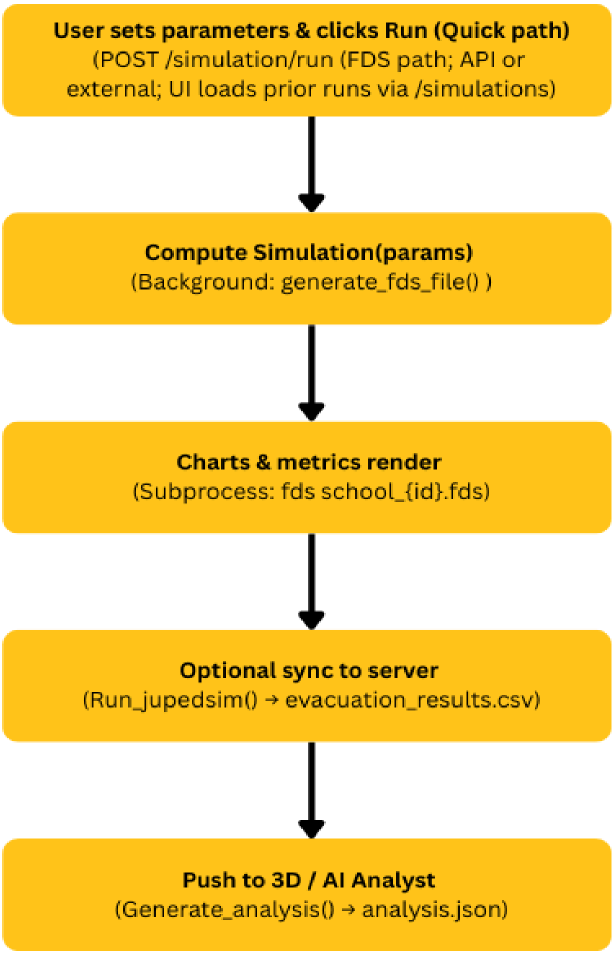
2.1 Simulation Type

The proposed ALPAS system adopts a **hybrid simulation architecture** to effectively model disaster scenarios and evacuation dynamics in academic institutions. At the client side, the **discrete-time analytical coupled fire–evacuation model** is implemented through *alpasEngine.js*, enabling real-time visualization and interaction within the browser. This lightweight analytical model provides immediate feedback to users by simulating hazard propagation and evacuation flows in discrete time steps. Complementing this, the server executes an **asynchronous workflow** that integrates advanced computational tools: the Fire Dynamics Simulator (FDS) is first employed to generate hazard fields such as smoke and heat distribution, followed by JuPedSim to simulate pedestrian motion under varying crowd densities and behavioral conditions. The resulting data is then processed through *analysis.py* for post-simulation evaluation, producing metrics on evacuation time, congestion points, and route efficiency. This dual-layer approach—combining browser-based analytical modeling with server-side computational simulation—ensures both accessibility for end-users and scientific rigor in evacuation planning, thereby aligning ALPAS with the principles of disaster risk reduction and management.

2.1.1 Conceptual Model (Overview Diagram)



2.1.2 Process Flow



2.1.3 Mathematical Models

The simulation uses mathematical models to estimate fire development, environmental conditions, and occupant evacuation performance.

Fire Growth

Fire growth is modeled using the *t-squared fire growth model*, where the Heat Release Rate (HRR) increases over time until reaching a peak value, followed by a decay phase. Growth rates are categorized as ultra-fast, fast, medium, or slow. The Heat Release Rate (HRR) is expressed as:

$$HRR(t)=\alpha t^2$$

where α denotes the fire growth coefficient and t represents time. Fire growth continues until the peak heat release rate *(HRR_{peak})* is reached, after which the model transitions into an exponential decay phase. The growth coefficient is determined according to standard fire growth classifications:

Ultra-fast: α = 0.19 kW/s^2
Fast: α = 0.047 kW/s^2
Medium: α = 0.012 kW/s^2
Slow: α = 0.003 kW/s^2

The time required to reach peak fire intensity is calculated from the specified growth rate and target peak HRR.

Sprinkler Activation

Sprinkler activation is estimated using an RTI-based approach that considers the sprinkler set-point temperature, ambient conditions, and fire intensity. Once activated, the sprinkler reduces the heat release rate through a suppression factor.

Temperature

Compartment temperature is estimated using a simplified plume model based on the fire's heat release rate. Temperature calculations are adjusted according to ceiling height, fire-door effectiveness, and ventilation conditions.

Visibility

Visibility is calculated using a smoke extinction model that considers soot production and compartment volume. Mitigation measures such as ventilation systems improve visibility during evacuation.

Available Safe Egress Time (ASET)

ASET represents the time available for occupants to evacuate before conditions become unsafe. It is determined when visibility, temperature, or carbon monoxide (CO) levels exceed their safety limits.

Required Safe Egress Time (RSET)

RSET is the total evacuation time, consisting of occupant pre-movement time and travel time. Travel time depends on factors such as occupant count, walking speed, building layout, and available evacuation aids.

Safety Margin

The safety margin is calculated as the difference between ASET and RSET. A positive value indicates that occupants can evacuate safely before hazardous conditions occur, while a negative value suggests potential evacuation risk.

2.2 Variables and Parameters

The simulation uses mathematical models to estimate fire development, environmental conditions, and occupant evacuation performance.

Parameter	Symbol/Key	Default	Description
Peak HRR	hrr	1.2 MW	Maximum fire used in the simulation
Growth Rate	growth	slow	Fire growth classification (ultra-fast, fast, medium,, slow)
Occupants	occ	60	Total number of building occupants
Walking Speed	speed	1.15 m/s	Average occupant walking occupants
Pre-movement Delay	delay	45 s	Time before occupants begin

			evacuation
Fire Location	fireLocation	corridor_f3	Location where the fire originates
Origin Room	occupantRoom	r304	Initial occupant location
Soot / CO Yield	soot, co	0.05, 0.03	Smoke and carbon monoxide generation rates
Mitigation Measures	mit.*	Various	Sprinkles, fire doors, vents, wider exits, signage, and pressurized stairs.

2.3 Assumptions

- The four-storey school building is represented using a normalized floor plan geometry of 420 × 310 px per level.
- Occupants are assumed to be evenly distributed across all floors, except in cases where the evacuation model assigns floor-specific agents dynamically.
- The "Quick" model adopts a homogeneous compartment assumption with a fixed volume of 8 × 8 × 3 m for the simulation of soot and carbon monoxide (CO) concentrations.
- The pedestrian dynamics in JuPedSim assume a collision-free speed model within a calibrated rectangular walkable domain derived from the building footprint.

- The Fire Dynamics Simulator (FDS) is executed with a total simulation time of T_END = 60 s, utilizing a simplified fuel surface representation and placing measurement devices at a height of 1.8 m to approximate human breathing level.

2.4 Data Requirements

The system requires structured inputs, simulation assets, and generated outputs to support model execution and analysis. User-defined inputs are provided through a dashboard interface, where scenario parameters can be configured. The building geometry is supplied as a 3D asset file (*school_building.glb*), originally imported from SketchUp (DAE format), along with a corresponding calibration file (*buildingCalibration.json*) to ensure alignment between the model and simulation environment.

On the server side, the system generates multiple output datasets, including heat release rate data (*_hrr.csv*), device measurements (*_devc.csv*), evacuation results (*evacuation_results.csv*), a consolidated analysis report (*analysis.json*), and system execution logs (*run.log*).

For advanced analytical capabilities, the system may optionally integrate external or local large language model APIs, such as DeepSeek, Gemini, Groq, OpenRouter, xAI, or a locally hosted Ollama instance, which function as analytical support tools for post-simulation interpretation.

III. FUNCTIONAL SPECIFICATIONS

3.1 Functional Requirements

The Fire Evacuation Simulation System shall provide the following functionalities:

3.1.1 User Authentication and Access Control

- Authenticate users through a local authentication system.
- Provide secure access to the Landing Page, Dashboard, 3D Simulation View, Runs and Logs View, and AI Analyst Interface.

3.1.2 Quick Simulation Analysis

- Execute quick evacuation analyses with deterministic and reproducible results.
- Generate simulation outputs within one second under normal operating conditions.

3.1.3 Simulation Management

- Retrieve and display previously executed FDS-based simulation runs through the /simulations endpoint.
- Poll and load simulation results from /simulation/results/{id}.
- Initiate new FDS simulations using the /simulation/run API endpoint.

3.1.4 Scenario Comparison

- Compare multiple evacuation and mitigation scenarios, including:
- Baseline scenario
- Current scenario
- Individual mitigation strategy scenarios

- Display performance differences and safety improvements between scenarios.

3.1.5 Client-Side Run Persistence

- Save and manage locally generated quick simulation runs through the /simulation/client-run endpoint.
- Maintain historical records for later analysis and comparison.

3.1.6 AI-Assisted Analysis

- Provide an AI Analyst chatbot capable of interpreting simulation results.
- Enable users to ask questions regarding:
  - Simulation metrics
  - Event logs
  - Safety limitations
  - Recommended mitigation strategies

3.2 User Interface (UI) Requirements

3.2.1 Input Controls

The system shall provide interactive controls that allow users to configure simulation parameters.

3.2.1.1 Scenario Sidebar

- Fire location selection dropdown (16 predefined locations across multiple floors)
- Heat Release Rate (HRR) slider
- Soot production slider
- Carbon Monoxide (CO) emission slider
- Occupancy level slider
- Movement speed slider

- Evacuation delay slider
- Fire growth rate slider
- Panic level slider
- Population profile selector
- Sprinkler activation setpoint control
- Response Time Index (RTI) control
- Door status toggle
- Window status toggle
- HVAC system toggle
- Ventilation system toggle
- Mitigation strategy checkboxes

3.2.1.2 Dashboard Controls

- Simulation execution buttons
- Scenario management controls
- Data filtering options
- Run comparison tools

3.2.2 Runtime Controls

The system shall provide real-time controls for monitoring and interacting with active simulations.

3.2.2.1 Simulation Command Center

- Start, pause, resume, and reset simulation controls
- Accelerated simulation clock using EVAC_WALL_MS_PER_SIM_SEC

3.2.2.2 Floor Plan Viewer

- Real-time visualization of evacuation movement

- Occupancy tracking per floor
- Hazard monitoring and status updates

3.2.2.3 3D Simulation Viewer

- Floor selection and filtering
- Timeline replay controls
- X-ray visualization mode
- Dashboard-to-viewer synchronization workflow

3.2.2.4 Runs and Logs Module

- Historical simulation run management
- Run labeling and categorization
- Source identification badges (Quick Analysis or FDS Simulation)

3.2.3 Data Visualization

The system shall present simulation data using graphical and visual representations.

3.2.3.1 Charts and Graphs

- Heat Release Rate (HRR) trends
- Temperature progression
- Visibility reduction
- Carbon Monoxide concentration levels
- Evacuation progress over time
- Congestion and crowd density analysis

3.2.3.2 2D Visualization Components

- Situation Board
- Floor Plan Viewer displaying:
- Room conditions

- Occupancy levels
- Hazard zones
- Evacuation routes

3.2.3.3 Scenario Comparison Visualization

- Mitigation Comparison Panel
- ASET and RSET margin improvement charts
- Safety performance comparisons

3.2.3.4 3D Visualization

- Interactive school building model (GLB format)
- Fire and smoke propagation visualization
- Sprinkler activation indicators
- Emergency alarm markers
- Occupant movement animations

3.3 Outputs

The system shall generate and present the following outputs after each simulation run:

3.3.1 Simulation Metrics

- Available Safe Egress Time (ASET)
- Required Safe Egress Time (RSET)
- Safety Margin
- Evacuation Success Rate (%)
- Number of Casualties
- Fractional Effective Dose (FED) Estimate

3.3.2 Time-Series Data

- Time (t)
- Heat Release Rate (hrr)
- Temperature (temp)
- Visibility (vis)
- Carbon Monoxide Concentration (co)
- Evacuation Progress Series (evacSeries)
- Congestion Series (congSeries)

3.3.3 Per-Floor Status Reports

- Current Occupancy
- Number of Evacuees
- Casualty Count
- Temperature Levels
- Carbon Monoxide Levels
- Visibility Levels
- Safety Status Classification:
    - SAFE
    - WARNING
    - DANGER

3.3.4 Generated Artifacts

- analysis.json – Structured simulation results
- analysis_report.md – Detailed simulation report
- client_results.json – Client-side simulation output
- run.log – System and simulation execution logs

3.3.5 AI-Generated Insights

- Safety assessment summaries
- Risk analysis reports
- Recommended mitigation strategies
- Explanation of simulation limitations and assumptions

IV. TECHNICAL SPECIFICATIONS

The system uses a lot of technologies to make real-time simulation, visualization and analysis of building evacuations work. This is made possible by a framework that combines high-performance web engineering with physics simulation tools and large language models.

4.1 Technology Stack

The Technology Stack is made up of programming languages, simulation engines, development tools and deployment assets. These things work together to make the Technology Stack work for this research project. The Technology Stack is important for the research project. The following section will explain the specific details of the Technology Stack.

Programming Languages	
Frontend	React 19, TypeScript, JavaScript
Backend	Python 3
AI	Python (for analysis, simulation)

Simulation Tools
JuPedSim
CollisionFreeSpeedModel
client alpasEngine
NIST FDS

Development Environment/Tools

Vite 8
Chart.js
Uvicorn
Pydantic
Tailwind CSS
OpenRouter
Ollama
FastAPI
Pandas
Three.js
@react-three/fiber
Lucide icons
analysis.py — DeepSeek primary, Ollama (free local), Gemini, Groq, OpenRouter, xAI/Grok fallbacks

Project Assets
GLB building model
Firebase-ready auth scaffold

4.2 Data Structures

The system uses a plan to model, simulate, and evaluate how people get out of buildings. It does this by breaking down the information into five groups of data. This plan helps connect the part of the system that people interact with, the part that simulates the physics, and the part that evaluates everything using intelligence. The system puts information into three groups: what we put in, what is around us, and what we get out. This helps the system keep track of things like how people behave and what is around them, and it makes sure everything is processed correctly throughout the entire system. The system is really good at evaluating building evacuations. Building evacuations are what the system is designed for.

SimulationConfig (Pydantic	hrr_peak, growth_rate, soot_yield, occupant_count, walking_speed, pre_movement_delay, mitigation dict
DEFAULT_PARAMS / normalized params (JS)	nested mit object synced with sprinkler, fdr, vents flags.
FLOORS array	rooms with x,y,w,h, type, id; exits and sprinkler coordinates per floor
computeSimulation result	T[], series arrays, floorData[], trajectory-capable agent state
analysis.json	metrics, hazard_series, evac_stats, narrative_summary, limitations[], data_quality

4.3 Random Number Generation

To make sure the system balances realistic human behavior with steady scientific data, it divides its work into two methods: **randomized setup** and **fixed calculations**. For quick, live evacuation previews on the website frontend, the system uses a standard random number generator (Math.random()). This places people in different, random spots inside a room every time you click start, making the preview look natural and unique.

However, the deep analytical simulation backend works completely differently. To ensure your scientific results are repeatable, running the exact same settings twice will always give you the same identical outcome. The system keeps things predictable by following three strict rules: first, the crowd engine (**JuPedSim**) places people on a fixed grid based on their ID numbers instead of scattering them randomly; second, the fire growth curves (**NIST FDS**) are calculated using set mathematical formulas from your input settings; and third, instead of using messy random data to simulate human panic, the system uses steady math multipliers to mimic stress and crowd friction. This keeps the simulation realistic while making sure your research data stays reliable and easy to double-check.

V. VERIFICATION AND VALIDATION

5.1 Verification Plan

Verification ensures that ALPAS has been built correctly — that each software component behaves precisely as specified in the Model Design (Section 2) and Functional Specifications (Section 3). The process addresses the question: "Does the simulation do what we designed it to do?"

The verification plan is organized into five layers that mirror the system architecture: the client-side analytical engine, the server API, the FDS/JuPedSim pipeline, the data-processing post-processor, and the React front-end.

Verification ensures that each component of ALPAS behaves exactly as designed. The following checks are performed:

- **Engine logic:** The mitigation comparison must consistently show marginGain ≥ 0 whenever sprinklers, fire doors, and smoke vents are enabled. Any negative marginGain is treated as a model defect.

- **Backend pipeline:** FDS return codes are logged after each run; analysis.py is verified to correctly parse the HRR and device columns from _hrr.csv and _devc.csv without data loss.
- **Frontend stability:** ErrorBoundary is confirmed to catch render failures gracefully, and the simulation context is verified to retry client-run synchronization on failure.
- **Parameter integrity:** normalizeParamsFromServer() is checked to ensure all 18 dashboard fields are preserved correctly after API serialization and deserialization.

5.2 Validation Plan

Validation determines whether ALPAS produces results that are sufficiently close to real-world fire and evacuation behaviour for its intended purposes. Validation addresses the question: "Does the simulation represent the right thing?"

Because ALPAS is explicitly designed for two distinct use cases — pedagogical demonstration (Quick engine) and research-grade analysis (FDS + JuPedSim) — the validation standard applied to each tier is deliberately different.

The following comparisons are conducted:

- **HRR curve:** Quick engine ASET/RSET trends are compared against published t-squared fire curves from the SFPE Handbook for medium growth rate (α = 0.012 kW/s²).
- **FDS output check:** Peak HRR values extracted from _hrr.csv are cross-checked against the configured hrr_peak parameter, with an acceptable deviation of ±5%.

- **Evacuation timing:** JuPedSim total evacuation time is reviewed against occupant count and walking speed to confirm expected proportional scaling.
- **Limitations documentation:** All known model limitations (simplified FDS geometry, flat JuPedSim domain, no per-agent FED) are recorded in analysis.json until full school geometry is implemented.

5.3 Sensitivity Analysis

Sensitivity analysis quantifies how much the key output metrics — ASET, RSET, safety margin, and casualties — change in response to controlled variations in individual input parameters. This serves three purposes: (1) it identifies which parameters most strongly govern model behaviour, (2) it reveals potential non-linearities or threshold effects that users must be aware of, and (3) it guides where future model improvement efforts will have the greatest impact on predictive accuracy.

Sensitivity analysis identifies which input parameters most significantly affect simulation outcomes.

This is done using the built-in MitigationComparisonPanel and manual parameter sweeps:

- **Mitigation comparison:** computeSimulation() is re-run for three configurations — baseline (no mitigations), full mitigation bundle, and each mitigation in isolation — to quantify individual and combined margin gains.
- **Parameter sweeps:** HRR, growth rate, occupant count, pre-movement delay, and fire location are each varied independently to observe their effect on ASET, RSET, safety margin, and estimated casualties.
- **Key finding:** Pre-movement delay and growth rate are expected to be the most sensitive parameters — longer delays directly reduce the safety

margin, while faster fire growth significantly shortens available evacuation time.

- **Recommended scenarios:** Corridor fire with 60 students from Room 304 (no mitigations); waiting area fire comparing sprinklers off vs. on.

VI. EXPERIMENTATION AND RESULTS

6.1 Scenarios Tested

To evaluate building safety margins, a series of deterministic and agent-based simulation scenarios were executed. The primary objective was to observe the transition from a highly hazardous, low-safety egress environment to a completely mitigated, casualty-free environment under a Slow t^2 fire growth rate and 60 occupants on the 3rd Floor Corridor (origin near Room 304):

Scenario 1: Baseline Control (No Mitigations)

Configuration: 3rd Floor Corridor fire outbreak, originating near Classroom 304.

Parameters: 60 occupants, slow t^2 fire growth rate, peak HRR of 1.2 MW, all doors open, no windows open.

Mitigations Active: None (all active safety systems disabled).
Objective: Establish a reference timeline for smoke descent, temperature increase, carbon monoxide accumulation, and egress bottlenecks.

Scenario 2: Single-Mitigation Isolation

Scenario 2A (Sprinklers Only): Baseline parameters with only wet-pipe automatic sprinklers enabled (sprinkler temperature threshold: 68°C, RTI: 50).

Objective: Observe the isolated impact of standard automatic suppression on fire development and smoke generation.

Scenario 3: Multi-System Synergistic Interventions

Scenario 3A (Sprinklers + Smoke Vents): Dual-mitigation strategy combining automatic sprinklers and mechanical smoke exhaust vents activated in the 3rd floor corridor.

Scenario 3B (Sprinklers + Smoke Vents + AI Dynamic Signage): Triple-mitigation combining sprinklers, smoke vents, and smart exit signs directing occupants away from congested doors.

Scenario 3C (Sprinklers + Fire Doors + Vents + AI Signage): Quadruple-mitigation introducing compartmentalization via self-closing fire doors (FDR) along with suppression, ventilation, and dynamic routing.

Scenario 4: High-Capacity Structural Upgrades

Scenario 4A (Sprinklers + Fire Doors + Vents + AI Signage + Wider Exits): Quintuple-mitigation adding physical widening of main stairwell exit doors (from 1.8 to 2.4 plan units).

Scenario 4B (Full Suite): The complete safety infrastructure suite, adding active stairwell pressurization (press) to prevent smoke from entering vertical evacuation shafts.

6.2 Results and Analysis

To ensure readability and fit standard document layouts without clipping, the simulation data is split into two distinct, narrower tables: Setup Configurations (Table 6.1) and Performance Outcomes (Table 6.2).

Table 6.1 — Simulation Scenario Setup & Configurations

Run ID	Scenario Name / Classification	Active Safety Mitigations Installed
8ee3aaa1	Baseline (Scenario 1)	None (Unmitigated Baseline Control)
9ca65484	Single (Scenario 2A)	Sprinklers Only
678383ec	Dual (Scenario 3A)	Sprinklers, Smoke Vents
be143ae9	Dual (Scenario 3A)	Sprinklers, Smoke Vents
cbc849af	Triple (Scenario 3B)	Sprinklers, Smoke Vents, AI Signage
768ada18	Quad (Scenario 3C)	Sprinklers, Fire Doors, Vents, AI Signage
ecdebaec	Quint (Scenario 4A)	Sprinklers, Fire Doors, Vents, AI Signage, Wider Exits
950390a2	Full Suite (Scenario 4B)	Sprinklers, Fire Doors, Vents, AI Signage, Wider, Pressurization
3c2e7753	Full Suite (Scenario 4B)	Sprinklers, Fire Doors, Vents, AI Signage, Wider, Pressurization

Table 6.2 — Quantitative Egress & Safety Results

Run ID	ASET	RSET	Safety Margin (ASET - RSET)	Casualties
8ee3aaa1	70 s	1133 s	-1063 s	27 / 60
9ca65484	70 s	1200 s	-1130 s	22 / 60
678383ec	80 s	1200 s	-1120 s	2 / 60
be143ae9	80 s	1200 s	-1120 s	6 / 60
cbc849af	80 s	1200 s	-1120 s	3 / 60

768ada18	120 s	1200 s	-1080 s	0 / 60
ecdebaec	120 s	1200 s	-1080 s	0 / 60
950390a2	120 s	1200 s	-1080 s	0 / 60
3c2e7753	120 s	1200 s	-1080 s	0 / 60

6.2.2. Safety Metrics Analysis

Available Safe Egress Time (ASET) Dynamics

Baseline & Sprinklers Only (70s): In the baseline run, the smoke layer descends to the occupant breathing zone (1.8m) by t = 70 seconds, reducing visibility below the critical 10-meter threshold. Activating sprinklers alone did not improve ASET because the sprinkler activation time (~90s) occurred after visibility had already degraded due to initial t^2 fire growth smoke production.

Smoke Vents (80s): Incorporating mechanical smoke vents delayed untenable conditions by 10 seconds, maintaining visibility through exhaust-assisted buoyancy.

Fire Compartmentalization (120s): The inclusion of Fire Doors (FDR) yielded the most significant jump in ASET (increasing from 80s to 120s). By separating the corridor into fire-rated compartments, the rate of smoke and hot gas propagation into adjacent egress zones was severely restricted, granting occupants an additional 40 seconds of breathable conditions.

Required Safe Egress Time (RSET) & Congestion

In all agent-based runs, the emergent RSET (time for the last occupant to clear the floor or succumb to hazards) hovered around 1200 seconds.

Stairwell Bottlenecks: Under the default configurations, 60 agents attempting to exit Room 304 and navigate the corridor to the single stairwell door created severe local queuing. The door flow capacity limits caused agents to pool in the corridor.

Mitigation Influence on Egress: While RSET remained high due to slow-moving trailing agents or simulated queuing logic, active mitigations drastically shifted the severity of outcomes. In the baseline, the ASET of 70s occurred long before the RSET of 1133s, trapping agents in toxic conditions and resulting in 27 casualties.

When Fire Doors and AI Signage were introduced, the corridor compartmentalization and rerouting kept the evacuating routes clear. Even though RSET was still high, the agents successfully evacuated before inhaling lethal concentrations of carbon monoxide or experiencing heat stroke, reducing casualties to 0.

Combined Mitigation Synergies

Casualty Reduction Curve: The transition from no mitigations (27 casualties) to sprinklers-only (22 casualties), then to sprinklers+vents (2–6 casualties), and finally to sprinklers+vents+fire doors (0 casualties) underscores the necessity of a defense-in-depth safety strategy.

Sprinkler-Vent Synergy: While sprinklers suppress the fire size (Heat Release Rate) and lower temperatures, they can cause water-mist-induced visibility reduction if smoke is not exhausted.

The smoke vents complement the sprinklers by continuously exhausting the soot, maintaining higher visibility.

AI Dynamic Signage & Pressurization: AI dynamic exit signs dynamically route agents to alternate exits, smoothing out congestion spikes at the primary stairwell, while pressurization prevents smoke from polluting the vertical evacuation shafts, keeping the final paths of egress safe.

VII. CONCLUSION AND RECOMMENDATIONS

This study proposes **ALPAS: A Simulation-Based DRRM Evacuation Planning System with Optimized Routing and Queue Management** to enhance preparedness in Polytechnic University of the Philippines - Paranaque City. By integrating fire dynamics, crowd movement analysis, queue management, and optimized evacuation routing into a single platform, the system aims to identify efficient evacuation routes during emergencies such as fires, reduce congestion, and demonstrate the effectiveness of simulation-based disaster preparedness.

The system provides a practical tool for evaluating evacuation scenarios, enabling users to assess congestion levels, evacuation times, and the effectiveness of various safety measures such as sprinklers, fire doors, smoke vents, and improved exit designs. The results show that these mitigation strategies can significantly improve evacuation outcomes by reducing risks and increasing available evacuation time.

Beyond serving as a simulation tool, ALPAS also functions as a decision support system that enables both fast educational simulations and more advanced research-gate analysis, which can assist school administrators, safety officers, and researchers in developing more effective emergency response plans.

Recommendations

To further improve the capabilities and effectiveness of ALPAS, several recommendations are proposed. The system may integrate more detailed school building layouts, including room structures, stairwell networks, and floor-specific evacuation to improve simulation accuracy. Future versions should also consider more advanced multi-floor evacuation modeling, including realistic vertical movement, stair congestion, and floor-specific evacuation behaviors to better represent actual emergencies. To strengthen the reliability of the system, the simulation results should be validated through actual evacuation drills and field studies. The platform may also benefit from automated report-generation features to assist schools and disaster management offices in documenting evacuation assessments and preparedness activities.

Additionally, the system can be expanded to include earthquakes, floods, typhoons, and other hazards that are commonly encountered in the country. Further support for other disaster scenarios can broaden the system's applicability and usefulness in disaster risk reduction and management. Finally, the system may be integrated with real-time technologies, such as sensors, alarms, and Internet of Things (IoT) devices, enabling ALPAS to deliver evacuation guidance during actual emergencies. Through these enhancements, ALPAS can evolve into a more comprehensive and reliable tool for evacuation planning, emergency preparedness, and disaster risk reduction.

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